

Small molecule inhibition of the IL23R p19 interaction for psoriasis

Decision: pursue a gated discovery program for an oral, non-peptide inhibitor of the extracellular IL23R-p19 interaction. Prioritize the inducible p19 pocket and retain a receptor-side hotspot program as a second route. The disease mechanism is strongly validated; conventional small-molecule druggability and a viable oral exposure profile remain the central uncertainties.

IL-23 contains p19 and p40. Here, "IL23R-p19 inhibition" means physically preventing the p19-containing cytokine from engaging IL23R, whether the compound binds the cytokine or the receptor. A compound that inhibits TYK2, reduces IL-23 production, or suppresses a STAT3 reporter alone does not meet that definition.

Public evidence was assessed through **9 September 2026**. The intended product is chronic oral treatment for adult moderate-to-severe plaque psoriasis. The proposed target product profile (TPP) ends at preclinical candidate nomination and readiness planning for first-in-human studies; clinical goals are aspirations, not predicted outcomes.

Table 1 Evidence and program implications

Question	Assessment	Decision implication
Is the pathway validated in psoriasis?	High confidence: human genetics and therapeutic blockade.	Spend discovery effort on chemistry and exposure rather than re-proving the pathway.
Can oral receptor blockade work?	Yes for peptides: FDA-approved icotrokinra, March 2026. ^{15,16}	An oral product must differentiate beyond avoiding injections.
Is a p19 pocket experimentally observed?	Yes: peptide complex 8UUI; pocket reported in 2025. ^{10,11}	Use an ensemble containing peptide-bound and receptor-free conformations.
Is a non-peptide oral drug established?	No approved direct non-peptide blocker was identified in the reviewed sources.	Treat fragments and reporter hits as starting points requiring mechanistic confirmation.
What is the decisive early experiment?	Link direct binding to receptor displacement and selective human-cell inhibition.	Do not progress a series on docking or binding alone.

Recommended priority

The first investment should produce a reproducible, site-mapped non-peptide series with a plausible route to serum-active potency. Stop or redesign if compounds bind the induced pocket but cannot displace the receptor, if cellular activity tracks cytotoxicity or kinase inhibition, or if free exposure cannot cover the required potency within a feasible oral dose.

Evidence labels: Established = human intervention or convergent experimental evidence. Supported = published biochemical, cellular or structural evidence with limits. Proposed = discovery judgment or a numerical design target. No compound was experimentally tested for this report.

10. Lecomte et al., 2025. [Identification of an induced orthosteric pocket in IL23 A new avenue for ...](#)

11. RCSB PDB, accessed 9 September 2026. 8UUI X ray structure of interleukin 23 in complex with peptide 23 446

15. FDA, 2026. [Drug Trials Snapshots ICOTYDE Original approval 17 March 2026](#)

16. Janssen Biotech DailyMed, March 2026. [ICOTYDE icotrokinra US prescribing information](#)

Target biology and therapeutic benchmarks

IL23R recognizes p19 while IL12Rβ1 recognizes the shared p40 subunit. Receptor assembly activates JAK2/TYK2-associated signaling and STAT responses, supporting IL-17-family and IL-22 output in responsive immune cells. Selective extracellular blockade should preserve IL-12 signaling, providing a mechanistic discriminator from shared-p40 or downstream inhibition. ¹³

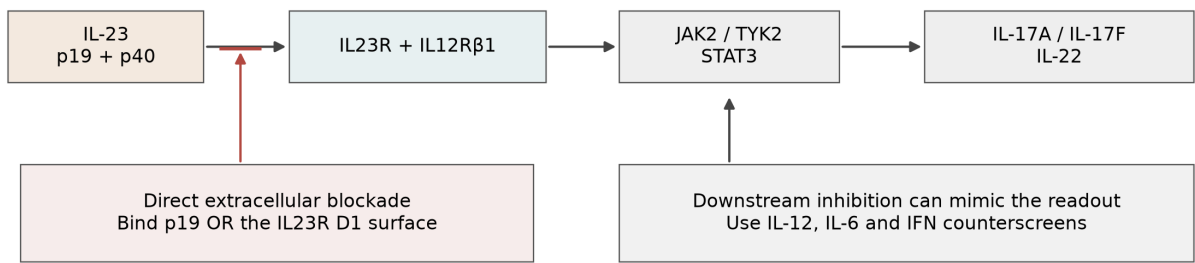


Figure 1 Mechanistic scope. Original schematic of the intended intervention and a major source of misleading assay activity. Signaling is simplified; the boxes do not depict molecular dimensions. ^{13,19}

Human genetics supports the direction of intervention. The protective IL23R Arg381Gln variant was associated with psoriasis protection; functional work found attenuated IL-23-induced Th17 effector responses. The variant is cytoplasmic, so it validates pathway attenuation rather than an extracellular binding pocket or a specific degree of pharmacological inhibition. ^{1,2}

Table 2 Relevant modality benchmarks

Benchmark	What it establishes	What it does not establish
p19 antibodies such as guselkumab	Clinical validation of selective cytokine blockade in psoriasis. ¹⁸	That the broad antibody epitope is a drug-sized pocket.
Icotrokinra oral IL23R peptide	Human efficacy and oral delivery of direct receptor antagonism; FDA approval 17 March 2026. ¹⁵	Non-peptide druggability or a safety advantage for a new chemotype.
Deucravacitinib oral TYK2 inhibitor	An oral therapeutic comparator acting downstream. ¹⁹	Direct blockade of IL23R binding to p19.
Proposed non-peptide inhibitor	Potential for a simpler oral formulation and distinct chemistry.	Clinical efficacy, low cost or superior safety before comparative evidence.

For a contemporary efficacy reference, FDA reports week-16 PASI-90 responses of 49.6% with icotrokinra versus 4.4% with placebo in PSO-3 (456 and 228 participants). PASI-90 means at least 90% improvement from baseline. These data calibrate ambition, not an animal-to-human conversion rule or a cross-trial ranking. ¹⁵

1. Capon et al., 2007. Sequence variants in the genes for the interleukin 23 receptor and its li...
2. Di Meglio et al., 2011. The IL23R R381Q gene variant protects against immune mediated diseases by...
13. Scientific Reports, 2024. JNJ 77242113 a highly potent selective peptide targeting the IL23 recepto...
15. FDA, 2026. Drug Trials Snapshots ICOTYDE Original approval 17 March 2026
18. Blauvelt et al., 2017. Guselkumab compared with adalimumab for moderate to severe psoriasis VOYA...
19. FDA, 2026. SOTYKTU deucravacitinib prescribing information

Public structural atlas

The core coordinate set supports both sides of the interface and multiple conformational states. All resolutions below are deposit-reported. Chain IDs use **author identifiers**; software using mmCIF label identifiers will differ for 5MXA and 8UUI. ^{4,5,7,9,11,22}

Table 3 Structures and appropriate uses

Entry and method	Composition and author chains	Use and material limitation
5MZV X-ray 2.80 Å Rfree 0.250	p40 A; p19 B; IL23R C; nanobody D	Native recognition geometry. Soluble receptor and crystallization nanobody; not a full membrane complex.
5MXA X-ray 2.501 Å Rfree 0.229	p40 A; p19 B	Receptor-free comparator. "Apo" here means no added receptor or peptide; p19 remains bound to p40.
6WDQ X-ray 3.40 Å Rfree 0.295	p40 A; p19 B; IL23R C; IL12Rβ1 D	Quaternary binding head. Only IL12Rβ1 D1 modeled; engineered receptor construct.
8OE4 Cryo-EM 3.60 Å	p40 A; p19 B; IL23R C; IL12Rβ1 D	Longer extracellular assembly context. Predimerization fusion constructs; flexible regions unresolved.
8UUI X-ray 2.43 Å Rfree 0.278	p40 A; p19 B; peptide 23-446 D	Induced-pocket template. Peptide is author D, label C; not a small-molecule complex.
9MFG X-ray 2.85 Å Rfree 0.246	IL23R A-D; VHH E-H	Alternative receptor-bound conformations and epitope context. A VHH complex, not icotrokinra.
6UIB X-ray structure	IL23 with peptide 23-652	Distal binding-site comparator and potential counterscreen reagent. ²⁶

Structure preparation priorities

Map chain sequences to the intended human constructs before selecting residues. Preserve disulfide bonds, inspect glycosylation and alternative conformations, and distinguish absent coordinates from absent sequence. Retain p40 during primary p19 modeling because it is part of the physiological cytokine. For receptor docking, compare isolated D1 with the longer extracellular domain and full-length cell-surface receptor.

Prepare receptor-free, receptor-bound and peptide-bound ensembles separately. Remove experimental partners only for a defined calculation; keep an annotated original. Review local density and side-chain support around any proposed ligand pose. The overall resolution and Rfree do not establish the accuracy of each pocket atom. Crystal packing, engineered constructs and unresolved loops limit claims about native dynamics.

4. RCSB PDB, accessed 9 September 2026. 5MZV IL23 IL23R Nb22E11 complex

5. RCSB PDB, accessed 9 September 2026. 5MXA Structure of unbound interleukin 23

7. RCSB PDB, accessed 9 September 2026. 6WDQ IL23 IL23R IL12Rβ1 signaling complex

9. RCSB PDB, accessed 9 September 2026. 8OE4 Predimerized human IL23 extracellular signaling complex

11. RCSB PDB, accessed 9 September 2026. 8UUI X ray structure of interleukin 23 in complex with peptide 23 446

22. RCSB PDB, accessed 9 September 2026. 9MFG Complex of IL23 receptor and VHH

26. RCSB PDB, accessed 9 September 2026. 6UIB Crystal structure of IL23 bound to peptide 23 652

Recognition interface and structural contacts

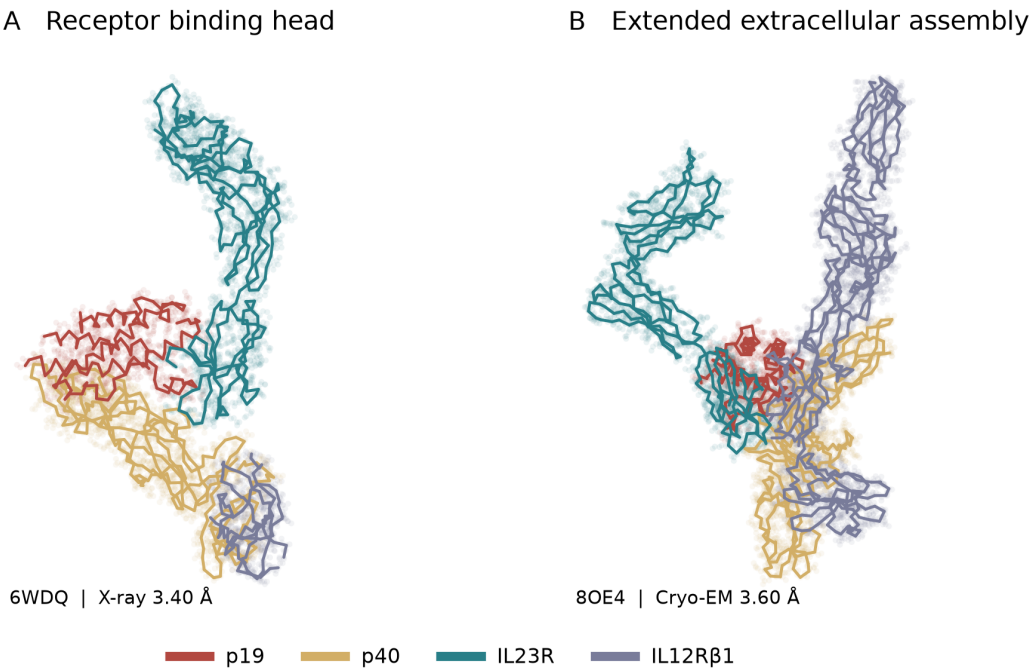


Figure 2 Original coordinate renderings. Cα traces and translucent heavy-atom projections of 6WDQ and 8OE4, with independently oriented views; not to a common scale. Glycans, waters and unresolved regions are omitted. 8OE4 was obtained using engineered predimerization constructs. No membrane or intracellular domains are shown.^{7,9}

The N-terminal IL23R immunoglobulin-like domain provides the p19 recognition surface. Human p19 Trp156 is a literature-supported functional hotspot; the corresponding mouse residue is Trp157. Mutational evidence makes this region worth prioritizing, but a residue contact is not itself a measurement of binding-energy contribution.^{3,29}

Table 4 Selected distances calculated from 5MZV

p19 atom	IL23R atom	Distance Å	Use
W156 CD1	D118 OD2	3.15	Receptor-side anchor region
W156 CH2	G116 N	3.34	Backbone-adjacent contact
W156 NE1	K117 N	3.58	Geometry, not an assigned hydrogen bond
W156 CB	S98 CB	3.62	Local packing contact
L160 CB	I28 CD1	3.91	Potential direction for ligand growth

Original analysis: the 4.0 Å heavy-atom cutoff identifies 31 residue pairs, involving 15 p19 and 18 IL23R residues in 5MZV. These are contact counts, not interface area or druggability scores. Distances are rounded to 0.01 Å for reproducibility and should not be interpreted as experimental precision. The supporting files contain the full contact list and method.

3. Bloch et al., 2018. Structural activation of pro inflammatory human cytokine IL23 by cognate ...
4. RCSB PDB, accessed 9 September 2026. 5MZV IL23 IL23R Nb22E11 complex
6. Glassman et al., 2021. Structural basis for IL12 and IL23 receptor sharing reveals a gateway for...
7. RCSB PDB, accessed 9 September 2026. 6WDQ IL23 IL23R IL12Rb1 signaling complex
8. Bloch et al., 2024. Structures of complete extracellular receptor assemblies mediated by IL12...
9. RCSB PDB, accessed 9 September 2026. 8OE4 Predimerized human IL23 extracellular signaling complex
29. Journal of Biological Chemistry, 2020. Deciphering site 3 interactions of interleukin 12 and interleukin 23 with...

The induced p19 pocket

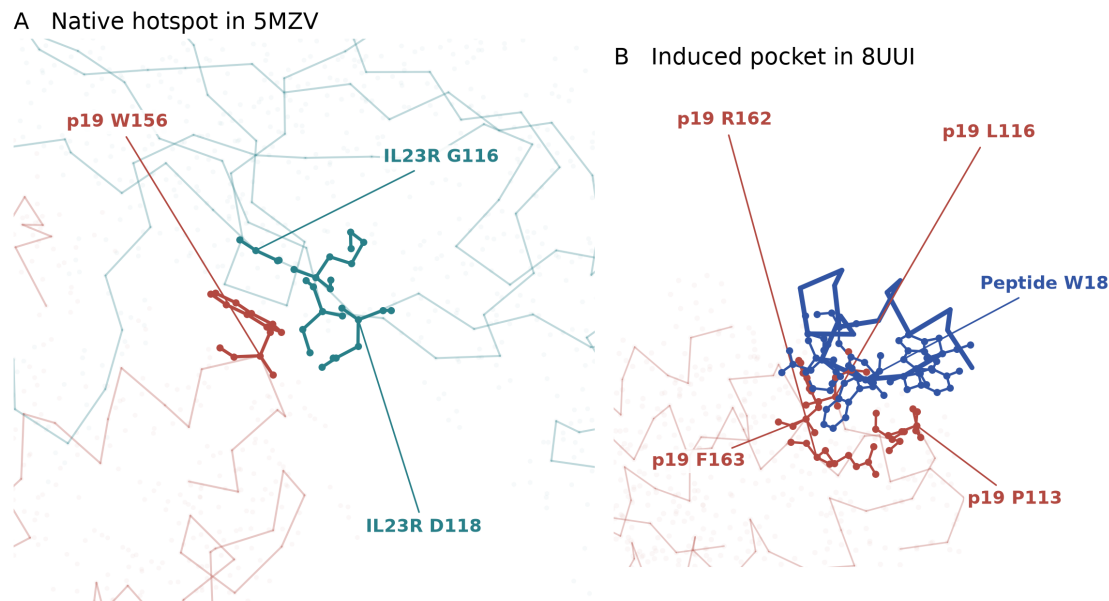


Figure 3 Original local coordinate views. A: native p19 W156 and nearby receptor residues in 5MZV. B: selected p19 and peptide side chains in 8UUI. Red is p19, teal is IL23R, blue is peptide. Atom projections illustrate geometry; they are not solvent-accessible surface or electrostatic calculations.^{4,11}

Lecomte and colleagues reported a peptide-induced pocket offset from the native receptor interface. Part of peptide 23-446 supplies steric blockade. Optimized peptides 14 and 15 achieved SPR K_d values of 16 and 20 nM and interaction-inhibition IC₅₀ values of 64 and 28 nM, respectively. The reported non-peptide fragment binders were much weaker, K_d 180 μM to 1.95 mM.¹⁰

Table 5 Coordinates suggest testable interaction vectors

8UUI contact	Calculated minimum Å	Proposed chemistry hypothesis
p19 R162 to peptide W18	3.60	A compact aromatic anchor near the guanidinium group.
p19 F163 to peptide F14	3.50	A second hydrophobic vector; avoid simply increasing lipophilicity.
p19 L116 N to peptide N17 O	3.17	A preorganized hydrogen-bond acceptor orientation.
p19 S114 O to peptide Y19 N	2.75	A donor vector, if desolvation and permeability remain acceptable.
p19 P113 to peptide W20	3.48	Growth toward the pocket rim and receptor-excluding volume.

Proposed interpretation: preserve two or more favorable vectors in a rigid, non-peptide core, then measure whether growth increases receptor displacement as well as affinity. A ligand that fills the pocket but leaves the receptor approach unobstructed may be pharmacologically silent. Do not assume that the peptide can be shortened into an inhibitor without losing its steric mechanism.

4. RCSB PDB, accessed 9 September 2026. 5MZV IL23 IL23R Nb22E11 complex
10. Lecomte et al., 2025. Identification of an induced orthosteric pocket in IL23 A new avenue for ...
11. RCSB PDB, accessed 9 September 2026. 8UUI X ray structure of interleukin 23 in complex with peptide 23 446

Conformational strategy and ligand design

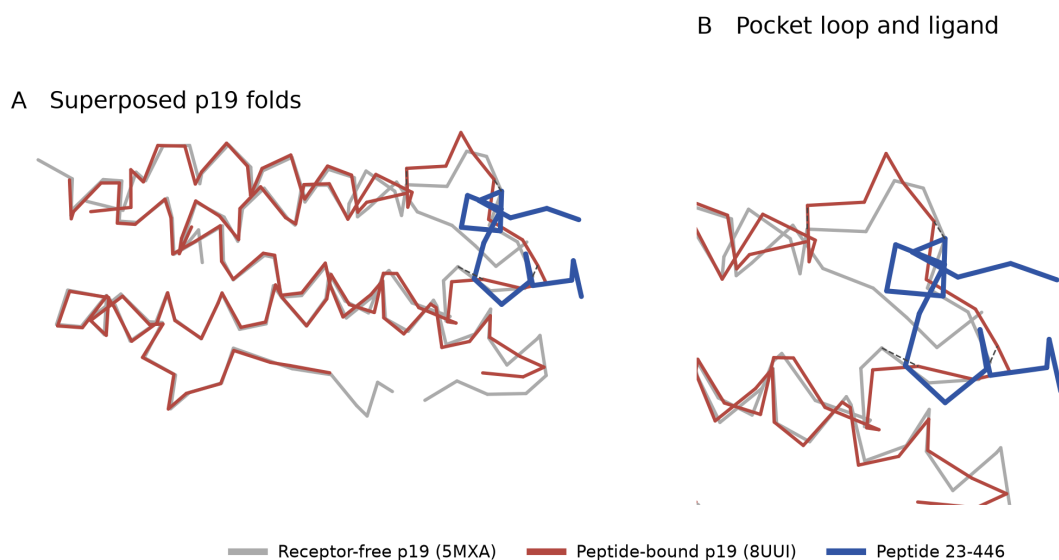


Figure 4 Original superposition of receptor-free 5MXA p19 and peptide-bound 8UUI p19. Gray and red are p19 C α traces; blue is peptide 23-446. A least-squares fit of 125 common C α atoms gives 1.28 Å RMSD. Dashed segments connect selected matched positions in residues 110-120; missing residues are not reconstructed. This is a static comparison, not a trajectory or a binding free-energy result.^{5,11}

Primary route through p19

Screen fragments against intact IL-23 and use the open pocket as one docking state. Combine label-free binding with competition and structure determination. Grow validated fragments toward the receptor-excluding region while tracking solubility, ligand efficiency and the relation between binding and inhibition. Small non-peptide macrocycles are a permissible backup if they improve shape control without forcing an impractical absorption profile. This is a proposed design route, not a demonstrated scaffold.

Receptor-side route

Seed a compact scaffold around the W156-recognition region on IL23R D1, then explore adjacent backbone and hydrophobic contacts. Macrocyclic dodecapeptide Compound C has HDX-supported receptor binding; its published pose was a docking hypothesis, not a ligand co-crystal.¹²

Receptor-binding VHHs and 9MFG provide additional epitope and conformational context, but neither constitutes a non-peptide ligand template.^{21,22}

Computational evidence boundary

A 2026 study identifies Inh-31 from a HEK-Blue IL23 reporter screen and proposes its mechanism using molecular dynamics. The accessible abstract and preview do not establish a validated drug-bound structure or a complete direct-binding and selectivity package. Treat it as a replication candidate, not proof that the protein interaction has been solved chemically.²⁰

Use simulations to rank testable poses and assess pocket persistence across replicas. A stable trajectory or favorable docking score cannot establish affinity, receptor displacement, selectivity or oral activity. Avoid a pose bridging a preassembled cytokine-receptor complex unless experiments distinguish antagonism from complex stabilization or agonism.

5. RCSB PDB, accessed 9 September 2026. 5MXA Structure of unbound interleukin 23

11. RCSB PDB, accessed 9 September 2026. 8UUI X ray structure of interleukin 23 in complex with peptide 23 446

12. ACS Medicinal Chemistry Letters, 2018. Deciphering binding interactions of IL23R with HDX MS Mapping protein and...

20. Manan et al., 2026. Deep learning guided discovery of IL23 IL23R macromolecular inhibitors An...

21. Ota et al., 2025. Engineering a protease stable oral single domain antibody to inhibit IL23...

22. RCSB PDB, accessed 9 September 2026. 9MFG Complex of IL23 receptor and VHH

Discovery strategy and assay cascade

Proposed program: run orthogonal fragment discovery and a compact, shape-diverse screening campaign, supported by peptide competition tools. Choose advancement criteria before screening. The p19 and receptor programs should use separate target-engagement assays but share functional and artifact controls.

Table 6 Strategy ranking

Priority	Approach	Advancement requirement and main risk
1	p19 fragment screening with structural follow-up	Two direct-binding methods and receptor displacement. Main risk: a real binder lacks sufficient blocking reach.
2	IL23R D1 hotspot screening	Binding persists with longer, glycosylated receptor and on cells. Main risk: a broad, shallow binding surface.
3	Peptide pharmacophore reduction or non-peptide macrocycles	Retain target selectivity while reducing polarity and improving exposure. Main risk: loss of cooperative contacts.
Exploratory	Allosteric p19 conformational trapping	Demonstrate inhibition without aggregation and map a mechanistic site. Main risk: pocket binding is uncoupled from signaling.
Low	Disrupt preformed p19-p40 assembly or covalent targeting	Require compelling structural and mechanistic evidence. Secreted IL-23 is disulfide linked; no validated selective covalent anchor is established here.

Table 7 Evidence needed to call a hit a direct inhibitor

Step	Readout	Essential controls
Identity and binding	Fresh synthesis or repurchase; LC-MS/NMR; SPR plus ligand-observed NMR or another independent method.	Solubility at test concentration, stoichiometry, reversibility, detergent sensitivity, protein/tag controls.
Interaction disruption	IL-23 versus IL23R binding in two formats with different detection principles.	Reverse immobilization, soluble competition, compound-only signal, receptor/cytokine titrations.
Native cellular function	Early IL-23-induced pSTAT3 followed by cytokine output in primary human cells.	Cell viability; receptor expression; IL-12/pSTAT4, IL-6/pSTAT3 and IFN α signaling.
Site and mechanism	Co-structure preferred; otherwise HDX-MS plus conservative site mutations and coherent SAR.	Verify mutant folding and baseline ligand binding; exclude nonspecific protein destabilization.

The 2025 pilot used peptide-based fluorescence polarization, a distal-site counterprobe and biophysical follow-up; one leading signal was judged false positive.^{10,26} This supports assay feasibility, but fluorescence interference and weak-fragment affinity make independent confirmation essential. DSF alone is not a sufficient binding proof.

Fit full concentration-response curves, report confidence intervals and assay concentrations, and repeat with independent compound preparations and donors. Competitive inhibition should respond coherently to ligand concentration and preincubation order. Ligand depletion and Fc avidity can distort apparent IC₅₀ or K_d; use monomeric reagents and equilibrium models where appropriate.

10. Lecomte et al., 2025. [Identification of an induced orthosteric pocket in IL23 A new avenue for ...](#)

12. ACS Medicinal Chemistry Letters, 2018. [Deciphering binding interactions of IL23R with HDX MS Mapping protein and...](#)

13. Scientific Reports, 2024. [JNJ 77242113 a highly potent selective peptide targeting the IL23 recepto...](#)

20. Manan et al., 2026. [Deep learning guided discovery of IL23 IL23R macromolecular inhibitors An...](#)

26. RCSB PDB, accessed 9 September 2026. [6UIB Crystal structure of IL23 bound to peptide 23 652](#)

Translation into psoriasis models

Proposed translational sequence: establish a human-cell exposure-response relationship, verify cross-species pharmacology, and then ask whether orally achieved exposure suppresses the same pathway in skin. Evidence that an oral peptide works in rats supports this sequence, but its cross-reactivity and pharmacokinetics cannot be assigned to a new small molecule. ^{13,14}

Table 8 Models and interpretation

System	Purpose and endpoints	Limit and decision
Primary human PBMCs and memory T cells	IL-23-induced pSTAT3; IL-17A/F or IL-22 under a validated stimulation context; viability.	Donor variability and low receptor expression require qualified responsive populations. Confirm in at least six donors before nomination.
Human whole blood	Nominal IC50/IC90, protein binding and ex vivo pharmacodynamic coverage.	Co-stimulation affects cytokine readouts. Match dilution and drug concentration to the assay matrix.
Psoriatic skin explants or immune-keratinocyte coculture	Pathway transcripts, cytokine secretion, epidermal response and tissue viability.	Limited lifespan and donor variation. A mechanistic bridge, not a substitute for clinical efficacy.
Human, rodent and nonrodent target panels	Direct binding and functional potency against matched cytokine/receptor pairs.	Select efficacy and toxicology species from data. Consider humanized models if a useful species is not responsive.
IL-23-induced skin inflammation	Exposure-linked reduction of thickness, epidermal hyperplasia and IL-17/IL-22-associated transcripts.	Acute cytokine challenge is not chronic human plaque psoriasis. Include treatment after inflammation begins.
Imiquimod model as supporting evidence	A second inflammatory context with pathway and histological measures.	Broad anti-inflammatory or toxic compounds can appear active. Do not use it alone to establish direct target mechanism.

IL-23 injection and imiquimod models reproduce selected psoriasis-like features and are established experimental tools. ²³ Use randomization, blinded assessment, prespecified endpoints, variance-based sample sizes, and an appropriate selective p19 or IL23R positive control. Benchmark agents must be active against the species used. Report both absolute disease measures and improvement relative to vehicle and the assay control.

Measure parent drug and relevant metabolites in plasma and skin, with attention to unbound concentrations. A high total skin concentration may reflect tissue binding or residual blood and can overstate exposure available to extracellular IL-23 or IL23R. Couple pharmacokinetics to early signaling and later skin responses rather than treating tissue accumulation as target engagement.

A useful translational biomarker panel includes stimulated blood pSTAT3, skin IL17A/IL17F/IL22 and DEFB4A, plus histology. Such pathway changes are supportive pharmacodynamics; the icotrokinra label explicitly leaves their relationship to clinical benefit incompletely understood. ¹⁶

13. Scientific Reports, 2024. JNJ 77242113 a highly potent selective peptide targeting the IL23 recepto...
14. Dermatology and Therapy, 2025. Translational pharmacokinetics of icotrokinra a targeted oral peptide tha...
16. Janssen Biotech DailyMed, March 2026. ICOTYDE icotrokinra US prescribing information
23. Current Protocols in Immunology, 2019. IL23 and imiquimod induced models of experimental psoriasis in mice

Target product profile for the intended therapeutic

All values in Tables 9 and 10 are **proposed planning targets**, not observed properties or regulatory acceptance thresholds. "Minimum" defines a credible product or nomination case; "preferred" defines differentiation. Clinical aspirations cannot be validated at the preclinical stage and must remain distinct from the preclinical criteria.

Table 9 Product concept and clinical aspirations

Attribute	Minimum credible profile	Preferred profile
Indication and population	Adults with moderate-to-severe plaque psoriasis eligible for systemic therapy.	Broad utility across patients seeking oral systemic therapy; later populations justified by separate data.
Mechanism	Direct, selective, reversible IL23R-p19 interaction inhibition by a non-peptide molecule.	Well-defined p19 or IL23R site, with preserved IL-12 and unrelated cytokine signaling.
Route and formulation	Oral solid dose; once daily or a clearly justified twice-daily regimen.	Once-daily tablet without fasting restrictions or complex administration.
Projected adult daily dose	Model-supported dose at or below 300 mg/day; feasibility is exposure dependent.	At or below 100 mg/day, with robust exposure under practical dosing conditions.
Efficacy aspiration	Week-16 PASI-90 near the approximately 50% oral direct-blockade benchmark, with patient benefit and durability.	PASI-90 of at least 60-70% at week 16 and improved convenience or tolerability; clinical testing required.
Time course and durability	Meaningful response during induction with maintained effect during chronic dosing.	Early itch improvement and sustained clearance with low interruption sensitivity.
Safety aspiration	Chronic-use benefit-risk compatible with psoriasis; manageable immune and organ risks.	Broad selectivity, minimal laboratory burden and low drug-interaction liability, all requiring clinical confirmation.
Manufacturing and access	Scalable synthesis, controlled stereochemistry and impurities, stable solid form.	Shorter economical synthesis, room-temperature shelf life of at least 24 months demonstrated by stability studies.
Differentiation	A material advantage beyond being oral: dose, food effect, formulation, access or tolerability.	Two or more advantages supported by evidence, without sacrificing efficacy.

The approved oral benchmark is demanding. Icotrokinra is a 13-amino-acid peptide, molecular weight 1898.17, dosed at 200 mg once daily with an empty-stomach instruction. Its label reports a 43% reduction in AUC with a high-fat meal.¹⁶ A food-flexible conventional small molecule is therefore a plausible product hypothesis; better tolerability or lower manufacturing cost cannot be assumed.

The efficacy aspiration is an analyst-selected development objective, informed by human target validation and current oral therapy.^{15,17,18} Preclinical inhibition percentages do not predict PASI responses. A topical program would require a different TPP focused on epidermal/dermal delivery, local tolerability and systemic spillover; it is not the lead concept assessed here.

15. FDA, 2026. [Drug Trials Snapshots ICOTYDE Original approval 17 March 2026](#)

16. Janssen Biotech DailyMed, March 2026. [ICOTYDE icotrokinra US prescribing information](#)

17. Bissonnette et al., 2025. [Oral icotrokinra for plaque psoriasis in adults and adolescents](#)

18. Blauvelt et al., 2017. [Guselkumab compared with adalimumab for moderate to severe psoriasis VOYA...](#)

Preclinical candidate profile

Proposed nomination gates: interpret every potency value in its stated assay matrix and against observed free exposure. Do not trade selectivity or safety for a single favorable potency number. Progression requires a coherent package, including an identified binding partner and an experimentally supported binding site.

Table 10 Quantitative candidate objectives

Attribute	Minimum objective	Preferred objective
Direct affinity and blockade	Kd ≤100 nM; biochemical PPI IC50 ≤100 nM; two independent assay principles.	Kd ≤10 nM and PPI IC50 ≤30 nM; ligand-bound structure.
Primary human-cell function	IL-23 pSTAT3 IC50 ≤100 nM; maximal inhibition ≥80%; meaningful effects across ≥6 donors.	IC50 ≤30 nM, maximal inhibition ≥90%, with matching later cytokine suppression.
Whole-blood potency	Nominal IC50 ≤300 nM and directly measured IC90 within the achievable exposure range.	Nominal IC50 ≤100 nM; modest matrix shift and reproducible ex vivo coverage.
Functional selectivity	≥30-fold IC50 separation versus IL-12, IL-6, IFNα and viability; test soluble concentration range.	≥100-fold separation and no material suppression at projected free clinical Cmax.
Exposure coverage	Unbound target-site trough ≥1× matrix-adjusted IC90, or validated time-dependent PK/PD rationale.	≥3× IC90 trough coverage with acceptable safety margins.
Oral PK	F ≥20% in a suitable species, or lower F with a convincing dose/exposure case.	F ≥40%; a predicted human profile supporting once-daily dosing.
Physicochemical profile	Prefer MW ≤650, cLogP approximately 1-4; solubility supports assays and projected dose.	MW ≤550; fewer exposed donors and low-to-moderate polarity consistent with measured absorption.
Efficacy package	Exposure-related activity in a target-relevant skin model plus a human ex vivo system.	≥70% suppression of the prespecified model disease increment and concordant pathway/histology effects.
Safety margin	Target ≥10-fold unbound AUC and Cmax separation at the relevant NOAEL from projected efficacious human exposure.	≥30-fold margins; no concerning organ or immune findings at covered exposures.
Early safety and DDI	Negative appropriate genotoxicity assays; acceptable safety pharmacology, CYP and transporter profile.	hERG IC50/free Cmax ≥30 as an initial screen, plus integrated cardiac assessment; low DDI potential.
Development readiness	Active, characterized material; scalable route; validated bioanalysis; species and formulation plan.	Reproducible batches, stable formulation, backup chemistry and ready IND-enabling study designs.

Numerical physicochemical ranges are preferences, not automatic exclusions; non-peptide macrocycles may justify exceptions. hERG and exposure ratios are screening targets, not proofs of cardiac or clinical safety. Toxicology duration and endpoints must match the intended clinical program.²⁴ The peptide benchmark has exceptional potency and low bioavailability; it does not set a transferable non-peptide absorption requirement.^{13,14}

13. Scientific Reports, 2024. JNJ 77242113 a highly potent selective peptide targeting the IL23 recepto...
14. Dermatology and Therapy, 2025. Translational pharmacokinetics of icotrokinra a targeted oral peptide tha...
24. FDA ICH, January 2010. M3 R2 Nonclinical safety studies for the conduct of human clinical trials...

Potency exposure and dose feasibility

A proposed extracellular inhibitor must achieve enough free drug where cytokine and receptor interact. Total plasma concentration, total skin concentration and biochemical K_d are insufficient on their own. Begin with a simple calculation, then replace assumptions with measured binding, distribution, clearance and pharmacodynamic data.

Illustrative calculation

For a unit-Hill, full-inhibition response, **IC₉₀ = 9 × IC₅₀**. Let K_{p,uu} represent unbound skin-interstitial to unbound plasma concentration, and f_{u,p} represent the unbound plasma fraction. Then a provisional coverage condition is **C_{min,total} ≥ IC_{90,u} / (f_{u,p} × K_{p,uu})**. This assumes target-site potency can be represented by an unbound cellular potency and that equilibrium approximations are adequate.

For illustration only: MW = 500 g/mol; f_{u,p} = 0.10; K_{p,uu} = 0.50; oral bioavailability F = 0.50; human clearance CL = 1 L/h; and average total plasma concentration = 2 × trough. The approximate daily dose is CL × C_{avg,total} × 24 / F after converting concentration to mg/L. None of these values is measured for a proposed compound.

Table 11 Sensitivity to unbound cellular potency

IC _{50,u} nM	IC _{90,u} nM	Total trough μM	Approximate dose mg/day
3	27	0.54	26
10	90	1.80	86
30	270	5.40	259

This calculation explains why a nanomolar biochemical hit can still fail the product profile. At the 30 nM row, doubling clearance or halving bioavailability doubles the estimated dose. A threefold preferred trough margin would triple it to approximately 778 mg/day under the same assumptions, exceeding the proposed product objective. Improving potency, free distribution, clearance or exposure shape must compensate.

How to replace the assumptions

Measure f_{u,p} and assay-medium binding over relevant concentrations, including concentration dependence. Estimate unbound skin distribution with a validated method; tissue-homogenate binding is an approximation and does not uniquely identify extracellular exposure. Build an oral absorption and disposition model incorporating solubility, permeability, first-pass metabolism and active metabolites.

For whole-blood assays, compare nominal blood concentrations to an exposure measure calibrated in the same matrix. Do not divide a whole-blood IC₉₀ by plasma f_u a second time. For tight-binding or time-dependent systems, use ligand depletion, receptor turnover and binding-kinetic models instead of the simple Hill approximation. Cytokine concentration can materially shift competitive potency.

Use the resulting model to project a dose range and uncertainty interval, then choose an exposure that is both pharmacologically active and compatible with safety margins. These calculations inform medicinal chemistry prioritization; they are not a proposed first human dose. A first-in-human starting dose requires the actual nonclinical safety and pharmacology package.

Safety risks and development decisions

Target selectivity reduces some off-target explanations but does not eliminate the consequences of IL-23 pathway suppression. Icotrokinra labeling includes infection and tuberculosis precautions and avoidance of live vaccines; fungal infections are among reported adverse reactions.¹⁶ A new non-peptide compound also introduces chemistry-specific metabolism, organ-toxicity and interaction liabilities.

Table 12 Risks and decisions

Risk	Evidence to resolve it	Stop or redesign trigger
Pocket binding without inhibition	Binding-site map, receptor displacement and functional SAR.	Affinity improves repeatedly while displacement and selective cell activity do not.
Reporter or aggregation artifact	Detection counterscreens, soluble concentrations, label-free assays and viability.	Signal depends on detector chemistry, detergent, insolubility or cell injury.
Downstream mechanism	Kinase profiling and IL-12/IL-6/IFN α pathway controls.	Cell potency tracks shared signaling inhibition rather than direct target engagement.
Insufficient oral exposure	Measured free PK/PD, dose solubility and projected human clearance.	Target coverage requires impractical dose or erodes safety margins.
Species mismatch	Matched species ligand/receptor pharmacology, exposure and tissue response.	In vivo efficacy is claimed in a species without demonstrated target activity.
Chronic immune or organ toxicity	Repeat-dose findings, immune endpoints, metabolite assessment and recovery.	Adverse findings overlap efficacious exposure without a plausible mitigation.
Limited product differentiation	Dose, food effect, formulation and efficacy hypotheses tested against current options.	Only advantage is oral delivery, which is already clinically available.

Staged funding and nomination

Gate 1: qualified reagents, independent hits and clean artifact controls. **Gate 2:** two tractable chemical series, direct interaction blockade, site evidence and selective primary-cell activity. **Gate 3:** serum-active potency, exposure-linked skin pharmacology and a plausible human oral dose. **Gate 4:** a candidate and backup with scalable chemistry, bioanalysis, safety margins and a justified IND-enabling plan. These are proposed milestones, not a forecast of elapsed development time.

For a conventional small molecule, use ICH M3(R2) to align nonclinical safety work with clinical duration and population. Include pharmacology, toxicokinetics, genotoxicity and relevant safety pharmacology; plan reproductive, longer-term and other studies at the appropriate stage. Consider accepted alternative methods and current agency guidance rather than assuming every historical animal study is mandatory.^{24,25}

Public polypeptide and lipidated-peptide applications show active intellectual-property coverage around IL23R inhibition.^{27,28} They identify relevant claim families to examine, not a freedom-to-operate conclusion. Before series commitment, assess composition, binding-site, use and assay claims for the actual chemistry and jurisdictions; the present review does not establish patent clearance.

16. Janssen Biotech DailyMed, March 2026. ICOTYDE icotrokinra US prescribing information

24. FDA ICH, January 2010. M3 R2 Nonclinical safety studies for the conduct of human clinical trials...

25. FDA CDER, accessed 9 September 2026. Streamlined nonclinical studies and acceptable new approach methodologies

27. Public patent application, 2023. US20230357323A1 Human IL23 receptor binding polypeptides

28. Public patent application, 2026. US20260000659A1 Lipidated peptide inhibitors of interleukin 23 receptor

Validation scope and reproducibility

What was checked

The main conclusions were checked against primary research, original regulatory materials and public structure deposits. The structural atlas was checked against downloaded mmCIF metadata for 5MZV, 5MXA, 6WDQ, 8UUI, 8OE4 and 9MFG. Coordinate-based contact tables and the p19 superposition were recomputed for this report. The biological conclusions are literature validation, not independent laboratory replication.

Icotrokinra status was resolved using the FDA approval date and prescribing information rather than older papers that called the agent investigational. The FDA announcement-date distinction matters: the regulatory snapshot gives 17 March 2026. The report uses one explicitly identified efficacy population for the numerical clinical benchmark, avoiding mixing rounded label tables with differently summarized trial tables.

Structural analysis method

Use model 1, author chain IDs and author residue numbering. Include standard protein residues and modeled non-hydrogen atoms; exclude solvent, glycans and symmetry mates from the contact calculation. Bio.PDB selects the higher-occupancy atom for disordered alternatives. For each interchain residue pair, report the minimum atom-to-atom Euclidean distance if it is ≤ 4.0 Å.

For the 5MXA-to-8UUI overlay, match p19 author-chain B residues by residue number and fit all 125 shared C α atoms using a least-squares rigid-body superposition. The resulting 1.28 Å RMSD is a global static fit; local motion depends on the selected alignment. Do not interpret missing coordinates as a modeled open channel or a calculated conformational probability.

The figures were generated from these coordinates as C α traces and atom projections, with selected side-chain bonds inferred from local atomic distances for display. They contain no invented docked drug, pocket volume, electrostatic potential or computed binding affinity. Projection can overlap atoms that are separated in depth; numerical contacts were computed in three dimensions.

Remaining evidence gaps

No independently reproduced non-peptide candidate package containing selective direct blockade, site-resolved binding, psoriasis-relevant oral PK/PD and chronic safety was established by the sources reviewed. This does not exclude undisclosed industry programs. The 2026 Inh-31 assessment is limited to accessible abstract/preview information; no potency number or experimentally proven binding pose is assigned to it.²⁰

The public literature search is targeted and decision-oriented, not a registered systematic review or an exhaustive patent landscape. Disease validation is high confidence. The ranking of structural approaches, all quantitative TPP gates and the dose scenarios are explicitly proposed analyses that require experimental testing.

Abbreviations

AUC, area under the concentration-time curve; DDI, drug-drug interaction; F, oral bioavailability; HDX-MS, hydrogen-deuterium exchange mass spectrometry; IC₅₀/IC₉₀, concentrations giving 50%/90% inhibition; K_d, equilibrium dissociation constant; NOAEL, no-observed-adverse-effect level; PASI, Psoriasis Area and Severity Index; PBMC, peripheral blood mononuclear cell; PK/PD, pharmacokinetics/pharmacodynamics; PPI, protein-protein interaction; SAR, structure-activity relationship; SPR, surface plasmon resonance; TPP, target product profile.

4. RCSB PDB, accessed 9 September 2026. [5MZV IL23 IL23R Nb22E11 complex](#)

5. RCSB PDB, accessed 9 September 2026. [5MXA Structure of unbound interleukin 23](#)

7. RCSB PDB, accessed 9 September 2026. [6WDQ IL23 IL23R IL12Rb1 signaling complex](#)

9. RCSB PDB, accessed 9 September 2026. [8OE4 Predimerized human IL23 extracellular signaling complex](#)

10. Lecomte et al., 2025. Identification of an induced orthosteric pocket in IL23 A new avenue for ...

11. RCSB PDB, accessed 9 September 2026. [8UUI X ray structure of interleukin 23 in complex with peptide 23 446](#)

20. Manan et al., 2026. [Deep learning guided discovery of IL23 IL23R macromolecular inhibitors An...](#)

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13. Scientific Reports (2024). JNJ 77242113 a highly potent selective peptide targeting the IL23 receptor provides robust IL23 pathway inhibition upon oral dosing in rats and humans.

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